

Bit and Parameter of the LLMs Model

7B model = 7 billion parameter model

A parameter is the learned weight within a neural network.

So, a 7B model in 16-bits means:

- The model has 7 billion parameters
- Each parameter is stored using the 16 bits (2 bytes)

--> 7 billion x 2 bytes = 14 GB

at's why people often say a 7B FP16 model needs around **14 GB VRAM**.

TERM	MEANING
7B	7 billion learned weights (parameters)
16-bit	Each weight uses 16 bits of precision
FP16	A common 16-bit floating-point format

Think of it like this:

- **Parameters** = how many numbers the AI learned
- **Bits** = how detailed each number is

Example:

A parameter might be stored as:

- 32-bit → very precise, more memory
- 16-bit → less precise, half the memory
- 8-bit → smaller/faster
- 4-bit → much smaller, some quality loss

So a:

- **7B FP16** model is large and higher precision
- **7B 4-bit** model is the same model compressed to use less memory

The intelligence mostly comes from the **7B parameters**, while the **bit size** affects efficiency and sometimes quality.